

# ROHAN GUPTA

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## EDUCATION

Birla Institute of Technology and Science, Pilani – Goa Campus

Oct 2020 – May 2024

Bachelor of Engineering in Computer Science

CGPA 8.89

## EXPERIENCE

Adobe | Software Development Engineer 2

Jul 2024 – Present

- Architected and led the migration of legacy InDesign plugins to a modern extensibility framework, designing public SDK APIs over an **interface-based, command-driven object model** with lock-free cross-thread data transfer, delivering **thread-safe**, cross-platform C++ components for InDesign's multi-threaded engine.
- Designed and delivered a **content-authenticity signing system** — using a **signal-responder export pipeline** and **command-based provenance persistence** — that embeds tamper-evident metadata certifying how AI content was created, exposed through SOAP APIs for automated server-side signing at scale, positioning the product for California SB-942 and EU AI Act compliance.
- Developed an **LLM-based AI agent** converting natural language into executable InDesign UI actions via structured JSON action planning, DOM-grounded reasoning with live application-state injection, and **RAG-based semantic retrieval** over UI documentation — achieving **~85% action-execution accuracy** with multi-turn, session-aware context.
- Built a **multimodal LLM pipeline** that uses structured output schema to convert flowchart images/text into editable diagram markup in ~15 seconds, with semantic graph analysis for intelligent theming and element-synchronized narration via parallel text-to-speech audio synthesis — **reducing manual diagram-creation time by ~90%**.
- Co-authored an internally approved corporate patent for an **AI-driven system using SAM-based segmentation and canny edge detection** to extract and vectorize text-fillable regions from raw images with **~92% region-detection accuracy**.

Flipkart | Software Development Intern

Jan 2024 – Jun 2024

- Redesigned Grocery platform product sorting algorithm for **optimal intra-category spacing**, driving higher click-through rates.
- Evaluated and benchmarked multiple text preprocessing technologies—including Apache Lucene, Jortho, local LLMs, and custom Trie data structures—to identify the optimal **low-latency** spell-correction approach for Flipkart's product categorization pipeline.
- Boosted cross-sell sales by **17%** with A/B-tested feature that auto-scrolls users to recommendation slots on adding items to cart.

Adobe | Product Intern

Jun 2023 – Aug 2023

- Drove **LLM integration into Adobe InDesign** — evaluated models via Adobe's Firefly gateway, designed a **structured-JSON response pipeline** that returns schema-enforced text variations with robust parsing, and integrated generated text into the native C++ document store, adopted for shipping GenAI feature suite.

North Eastern Space Application Centre | Research Intern

May 2022 – Jul 2022

- Built a satellite-driven water monitoring platform using **Random Forest classification** trained on 4 landcovers (water, vegetation, urban, bare) on Sentinel-1/2 imagery with **85.88% accuracy**, automating water-body area extraction across 24-month time-series for flood trend analysis in Northeast India, served via GeoServer and REST API backed by MongoDB.

Teaching Assistant for Data Structures and Algorithms | under Prof. Sujith Thomas

Jan 2023 – May 2023

- Designed lab contests and delivered DSA lectures for juniors, creating problem sets and explaining core algorithms.

## TECHNICAL SKILLS

**Languages:** C, C++, Java, Python, MySQL, JavaScript, TypeScript

**Technologies/Frameworks:** Kafka, Redis, Docker, Kubernetes, MongoDB, LangChain, LangGraph, Linux, Git

## PROJECTS

Real-Time Clinical AI Triage Pipeline | Kafka, Redis, MongoDB, Python, Ollama, Docker Compose

May 2026 - Present

- Engineered a **distributed real-time stream processing pipeline** leveraging Kafka partitioning for high-throughput ingestion (200 readings/sec) and distributed caching via Redis to filter sensor artifacts, driving a 70%+ reduction in false alarms.
- Architected a concurrent AI triage agent with 4 parallel Ollama LLM inferences and 3-retry **dead-letter fault tolerance**, monitored via Grafana dashboards tracking inference latency (P95 < 15s) and suppression rates across 10 patient streams.

Multi-Agent Architecture Review Tool | Python, LangGraph, Ollama, ChromaDB, Rich

Apr 2026 - Present

- Developed a multi-agent CLI utilizing a **LangGraph state machine** with 3 LLM agents evaluating system designs, leveraging **ChromaDB RAG for semantic information retrieval** via nomic embeddings and paragraph-aware chunking.
- Shipped a **deterministic evaluation harness** measuring cross-reference accuracy, persona consistency, and hallucination rates, achieving 87.5% on gold fixtures vs 29.7% on degraded, validating a **58-point regression signal**.

## ACHIEVEMENTS / EXTRACURRICULAR

- Received the **Adobe Spot Award** for exceptional engineering performance. Nov 2025
- Achieved a maximum rating of **1547** on [Codeforces](https://www.codeforces.com/); solved **750+** problems on [LeetCode](https://leetcode.com/). Present
- Dean's Merit List (GPA 9.67)** in second semester; highest score in **DSA (98/100)**. 2021 – 2022
- Core member of BITS Goa coding club (AlgoManiax) and astronomy club (SEDS Celestia). 2020 – 2023
- Won **2 Champions Tropies and 1 Gold Medal** in Brainobrain's National Abacus Competitions. 2012 – 2014